# NITROGEN BALANCE AND TRANSPORT

# Nitrogen Transport

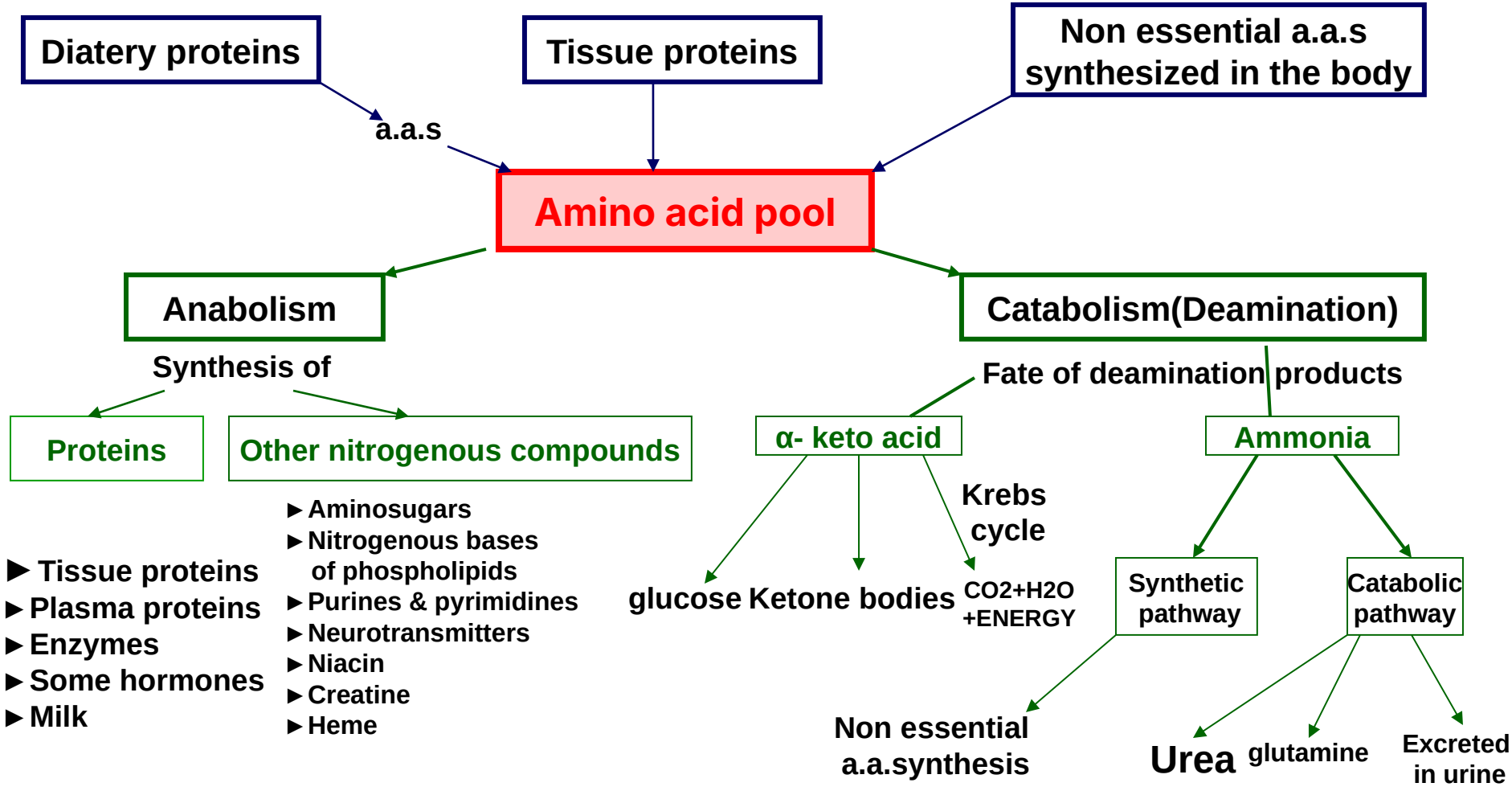
1. Discuss in general terms the use of amino acids for the synthesis of nitrogen-containing compounds
2. Discuss the various functions of glutamine
3. Discuss the roles of transamination
4. Discuss the glutamate dehydrogenase (GDH) & regulation
5. Discuss ammonia disposal
6. explain positive and negative nitrogen balance.

# Introduction

- Amino acids must be obtained from the diet, synthesized *de novo*, or produced from normal protein degradation
- Any amino acids in excess of the biosynthetic needs are rapidly degraded
- Catabolism of amino acids involves two phases:

1. The removal of the α-amino groups (by transamination and subsequent oxidative deamination) forming ammonia and the corresponding α-keto acid.
2. The carbon skeletons of the α-ketoacids are converted to intermediates of energy producing pathways

# ▶ SOURCES & FATE OF THE AA. POOL:



**Glutamate** plays the central role in mammalian nitrogen flow, serving as both a nitrogen donor and nitrogen acceptor.

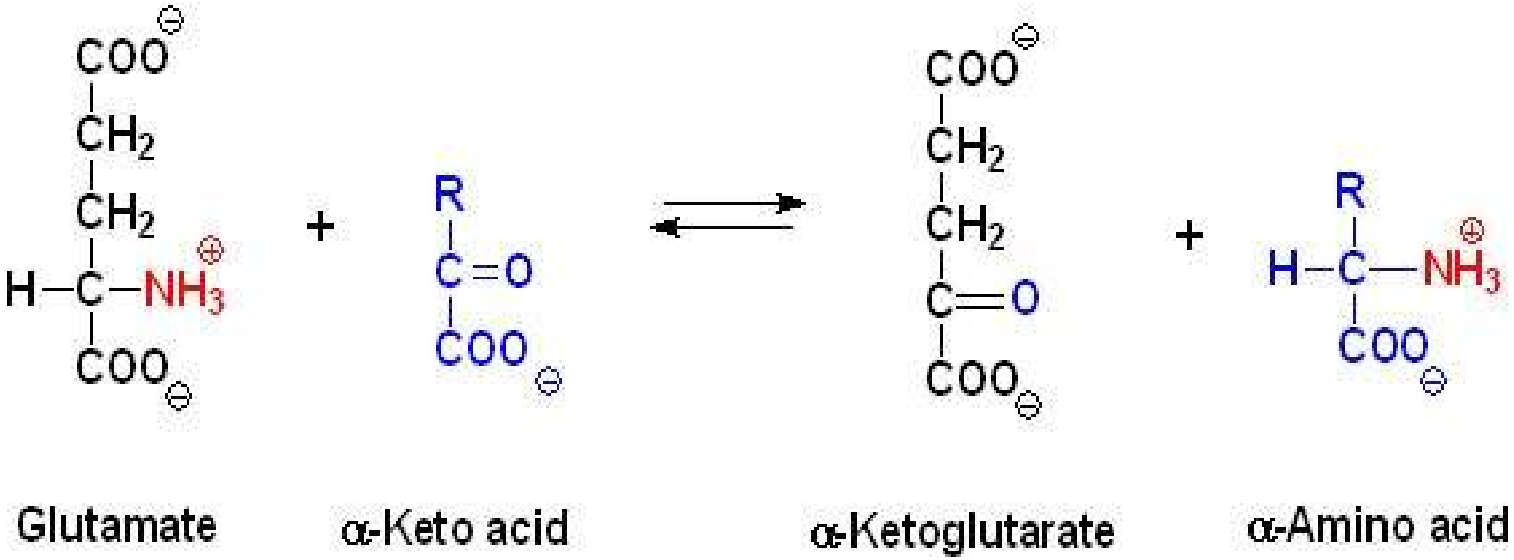
# Nitrogen Transport

- Reduced nitrogen enters the human body as
  - dietary free amino acids }
  - proteins } ammonia (by GIT bacteria).

- The two *principal* enzymes,
  - **Glutamate dehydrogenase and**
  - **Gutamine synthetase**, cause conversion of ammonia into the amino acids glutamate and glutamine respectively.

# Nitrogen Transport

- Amino and amide groups from these 2 substances are freely transferred to other carbon skeletons by transamination and transamidation reactions.

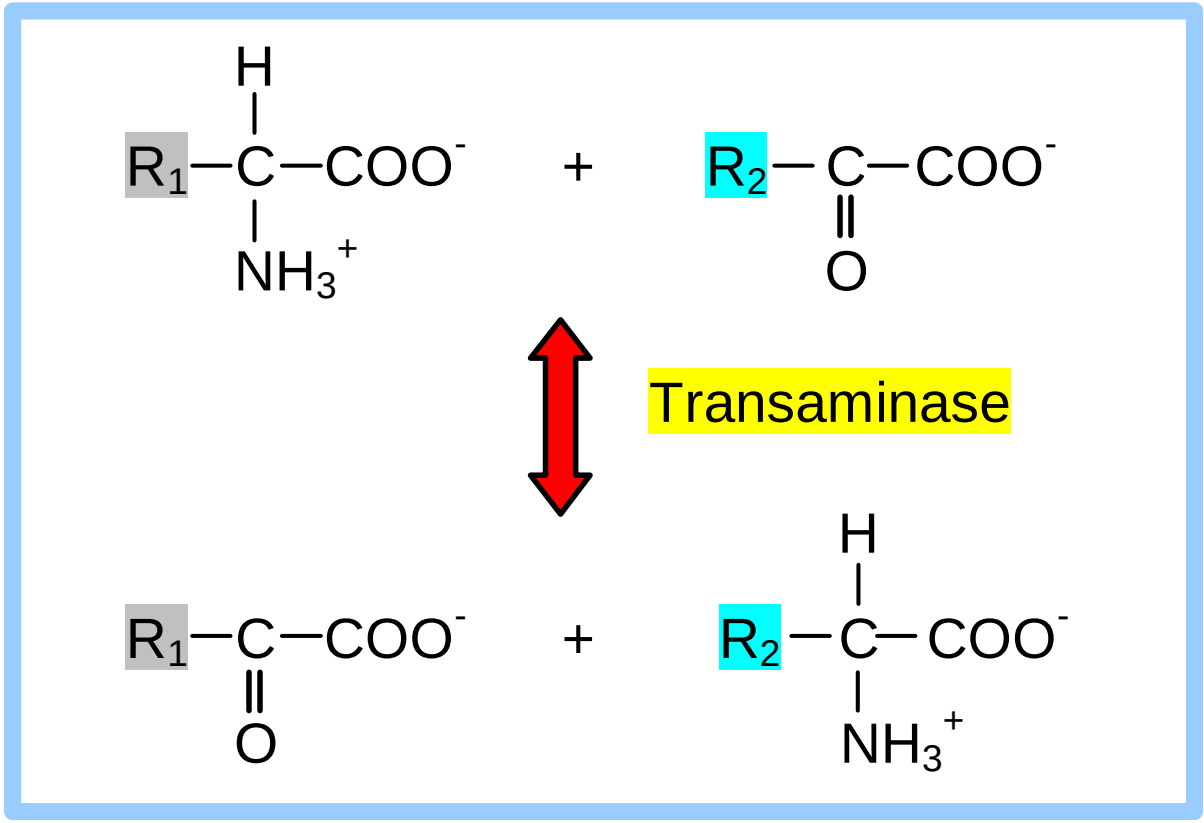


# Transamination Reactions

## Disposal of Amino Groups

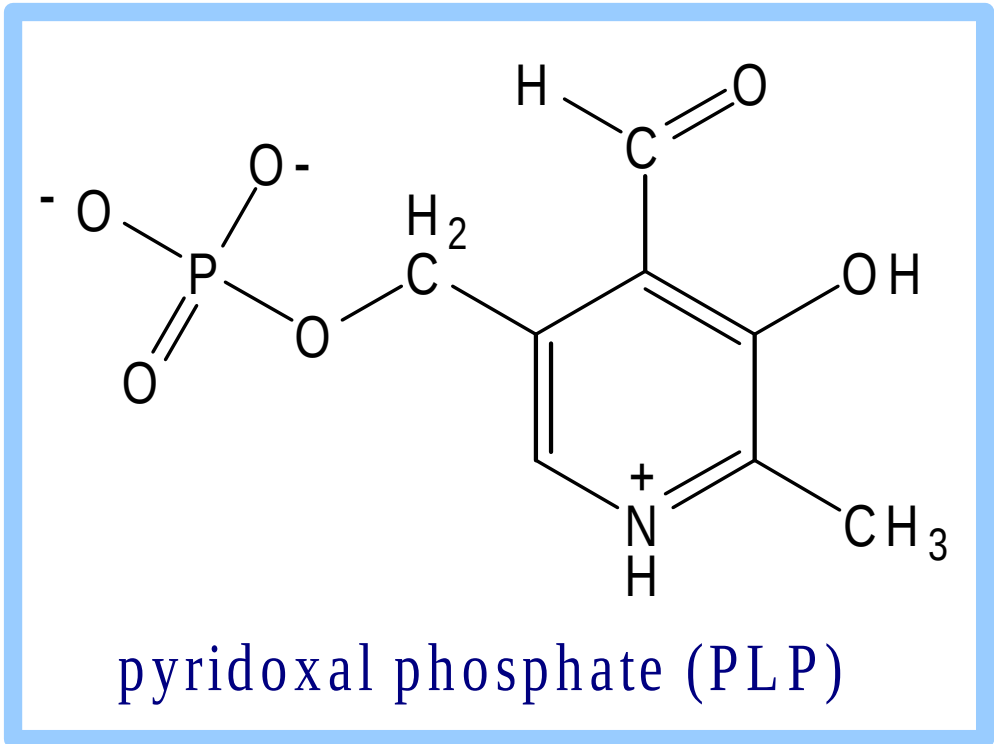
The most common compounds involved as a donor/acceptor pair in transamination reactions are **Glutamate and α-ketoglutarate**

**Transaminases** (aminotransferases) catalyze the reversible reaction at right.



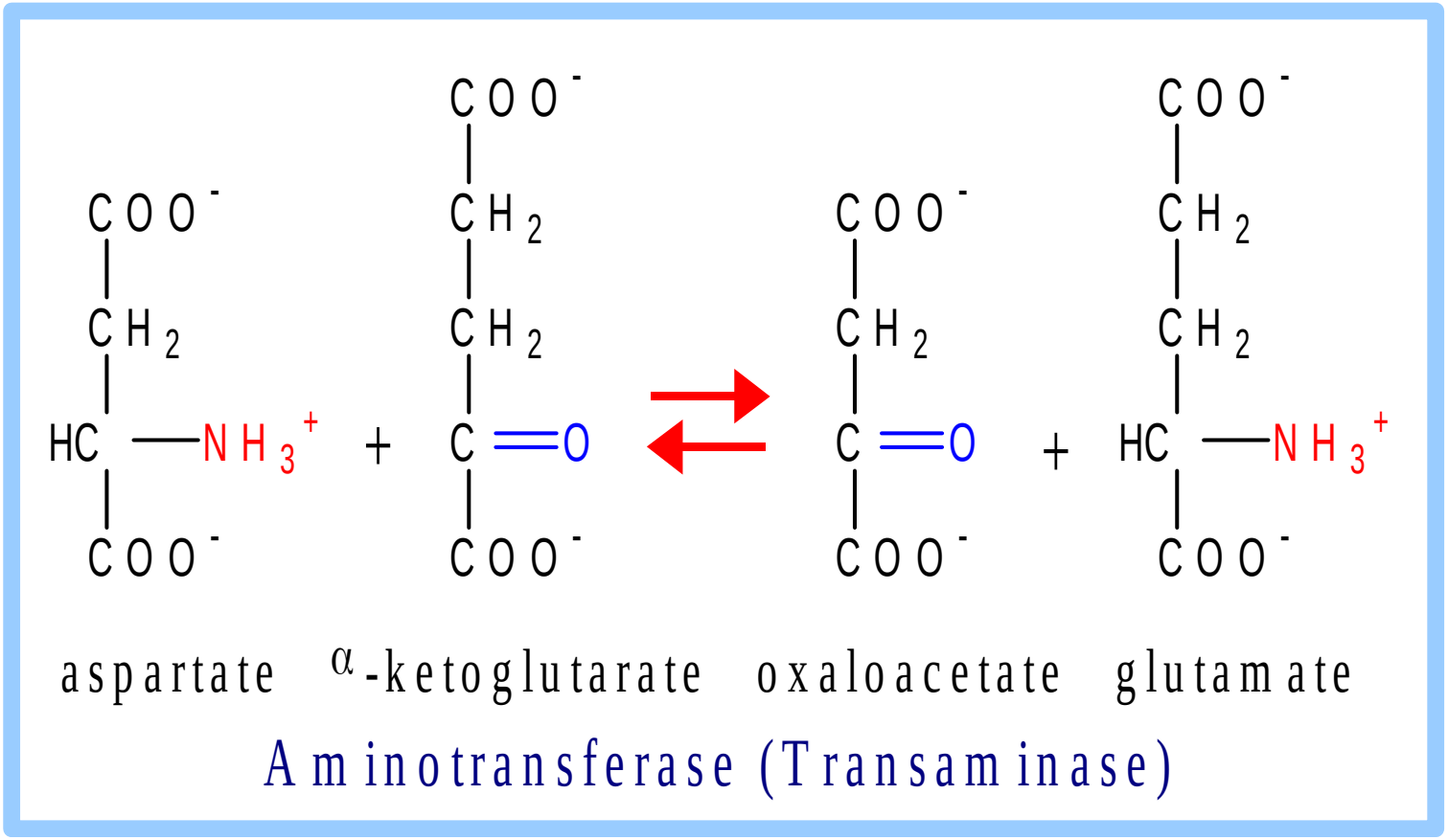

There are multiple transaminase enzymes which **vary in substrate specificity**.

Some show preference for particular **amino acids** or classes of amino acids as amino group donors, and/or for particular **α-keto acid acceptors**.



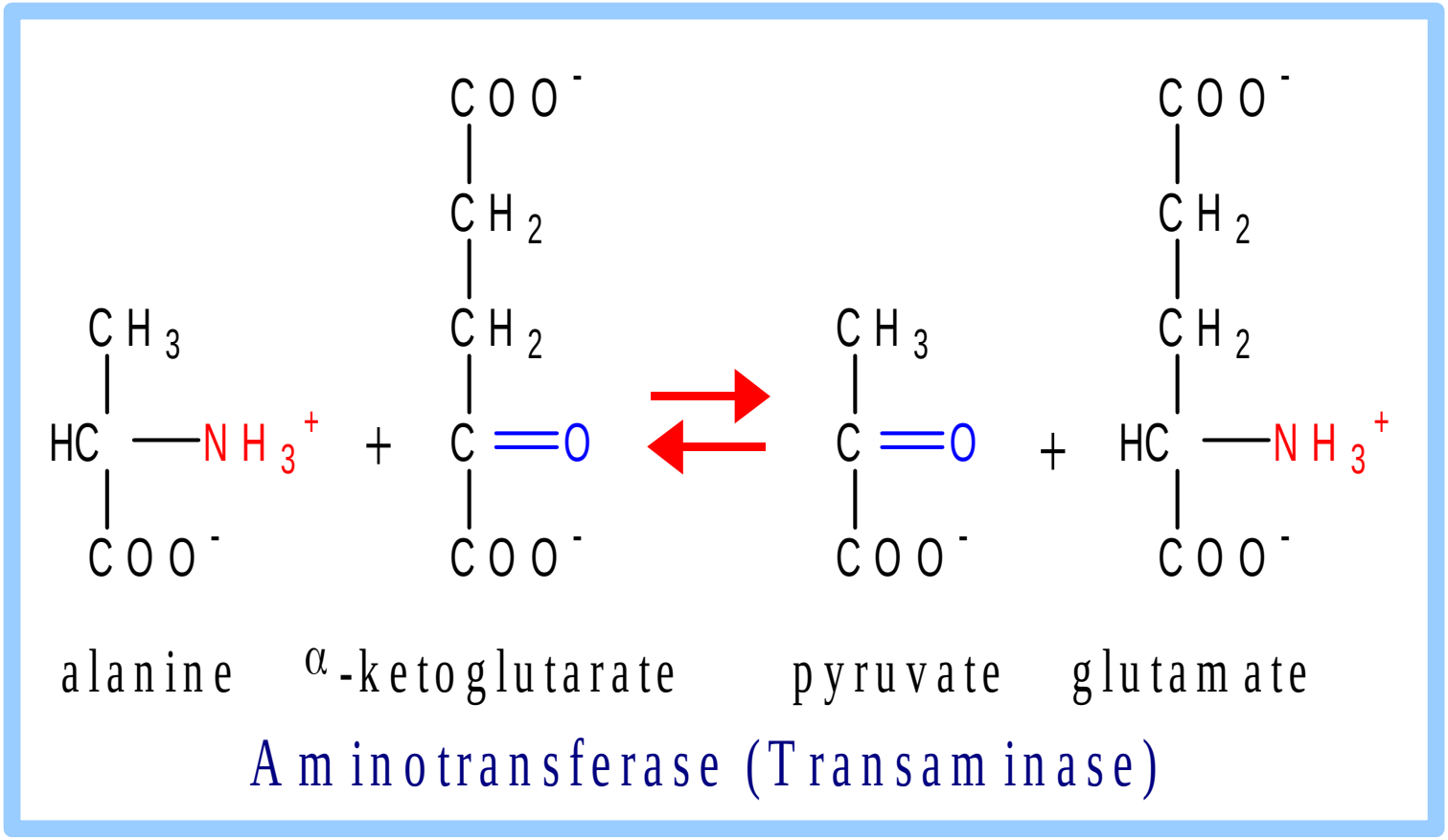

pyridoxal phosphate (PLP)

The prosthetic group of Transaminases is **pyridoxal phosphate** (**PLP**), a derivative of vitamin $B_6$.




Example of a **Transaminase** reaction:

- Aspartate donates its amino group, becoming the α-keto acid oxaloacetate.
- α-Ketoglutarate accepts the amino group, becoming the amino acid glutamate.

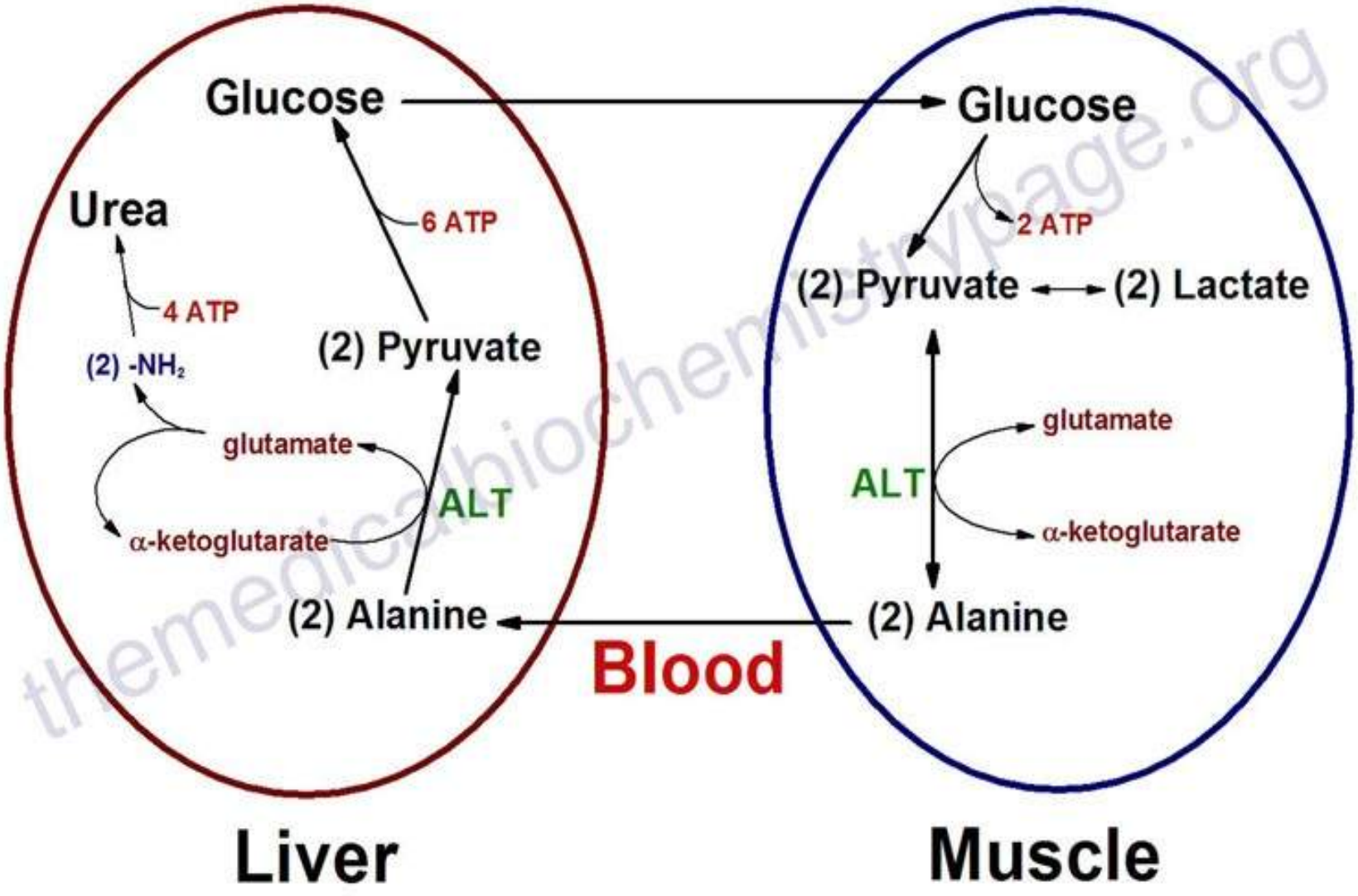



In another example, **alanine becomes pyruvate as the amino group is transferred to α-ketoglutarate**.

# Alanine transaminase

- Alanine transaminase has an important function in the delivery of skeletal muscle carbon and nitrogen (in the form of alanine) to the liver.
- In skeletal muscle, pyruvate is transaminated to alanine
- This is an additional route of nitrogen transport from muscle to liver referred as the glucose-alanine cycle

# Glucose-Alanine Cycle




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# Alanine transaminase, ALT.

- The glucose-alanine cycle is used primarily as a mechanism for skeletal muscle to eliminate nitrogen
- Glucose oxidation produces pyruvate which can undergo transamination to alanine catalyzed by Alanine transaminase, ALT.

•Transaminases **equilibrate amino groups** among available $\alpha$-keto acids.

**This permits synthesis of non-essential amino acids, using amino groups from other amino acids & carbon skeletons synthesized in a cell.**

•Although the amino N of one amino acid can be used to synthesize another amino acid, **N must be obtained in the diet as amino acids** (proteins).

# GENERAL STRATEGY

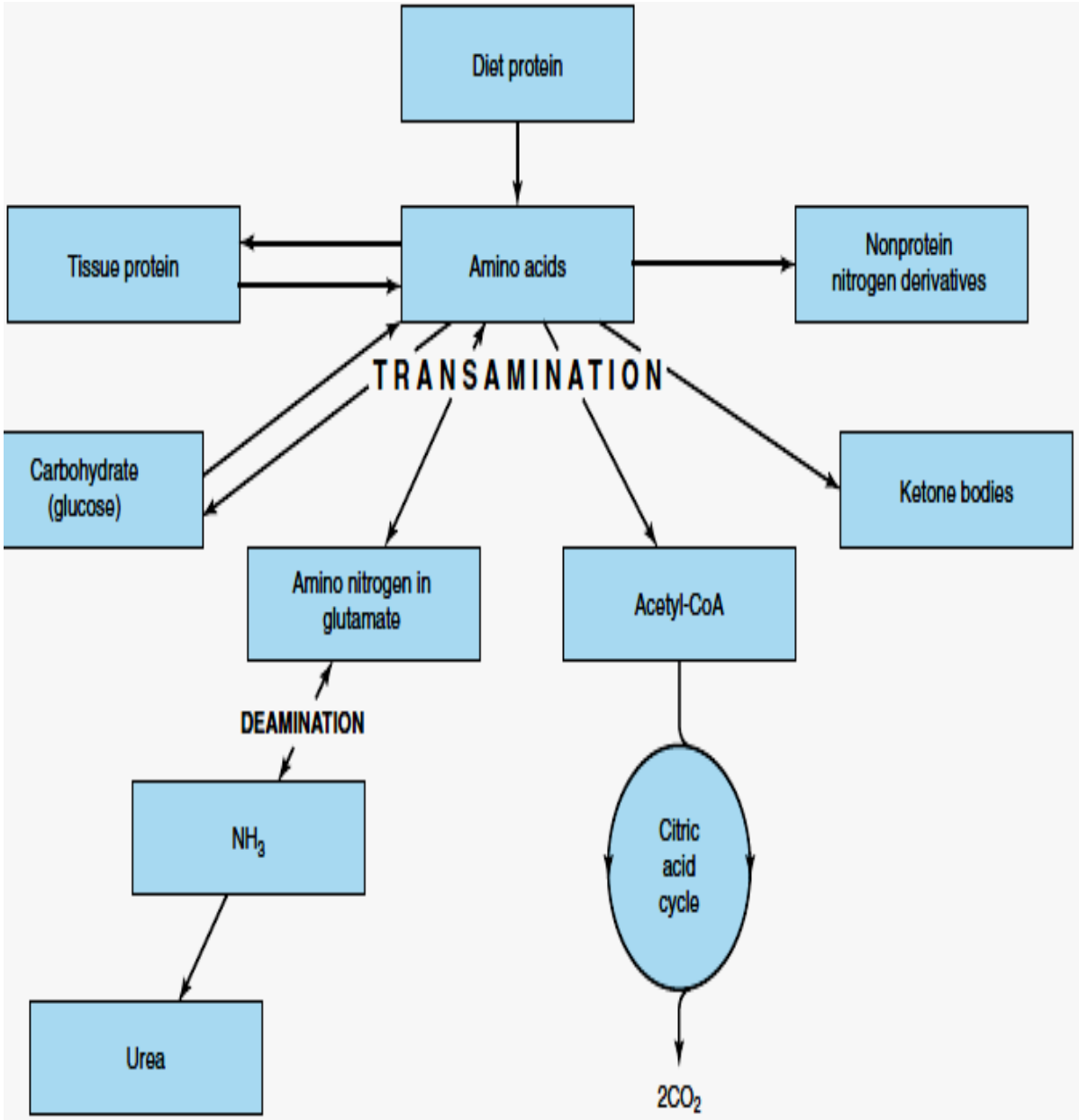
- Removal of N from amino acid by *transamination* (generally first or second step of amino acid catabolic pathways) and collection of N in glutamic acid
- *Deamination* of glutamic acid with release of $NH_4^+$ by glutamate dehydrogenase

Collection of N in glutamine or alanine for delivery to liver

- Removal of $NH_4^+$ by :

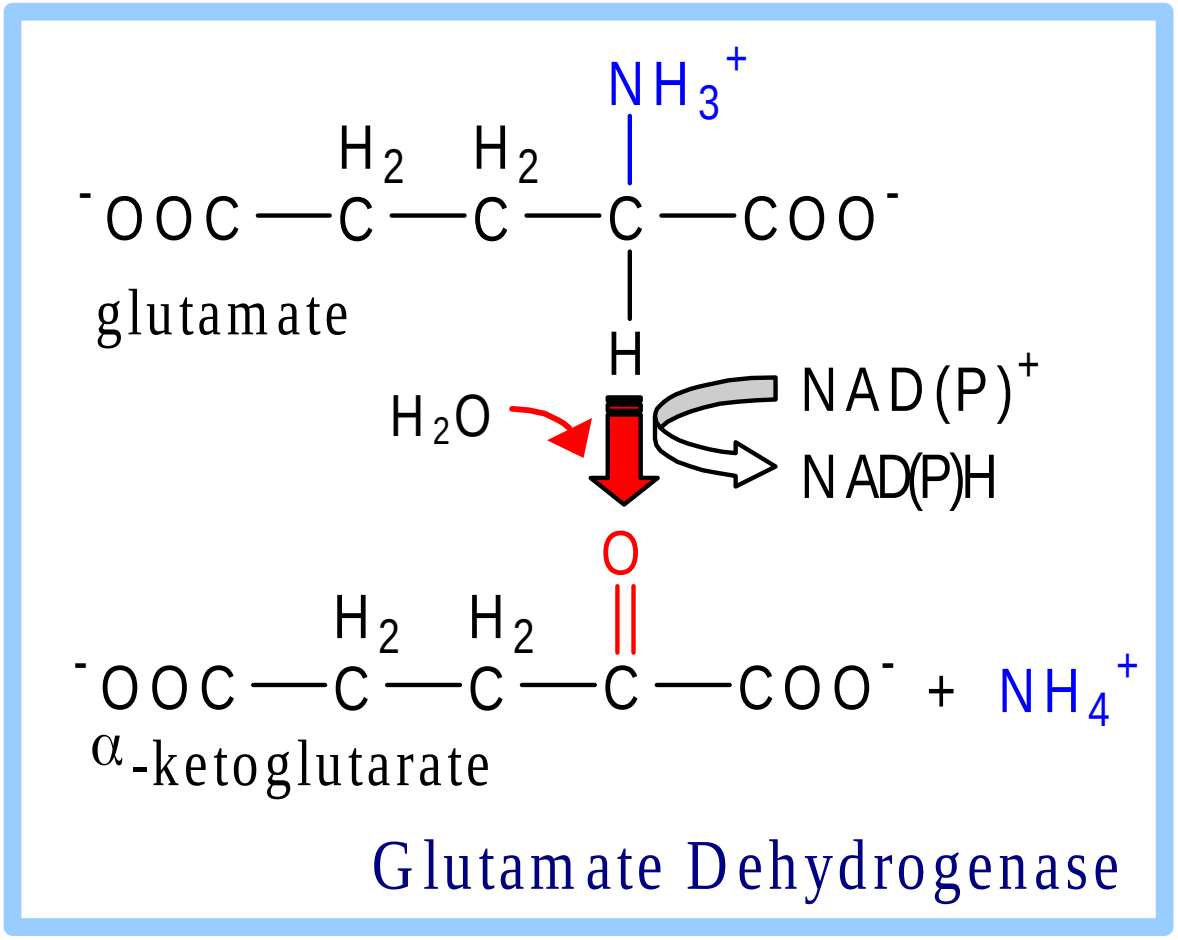
i. secretion; or

ii. Conversion to urea or other less toxic form.




ure 15–4. Overview of amino acid metabolism showing the major pathways and end products.

**Glutamate Dehydrogenase** catalyzes a major reaction that effects **net removal of N** from the amino acid pool.

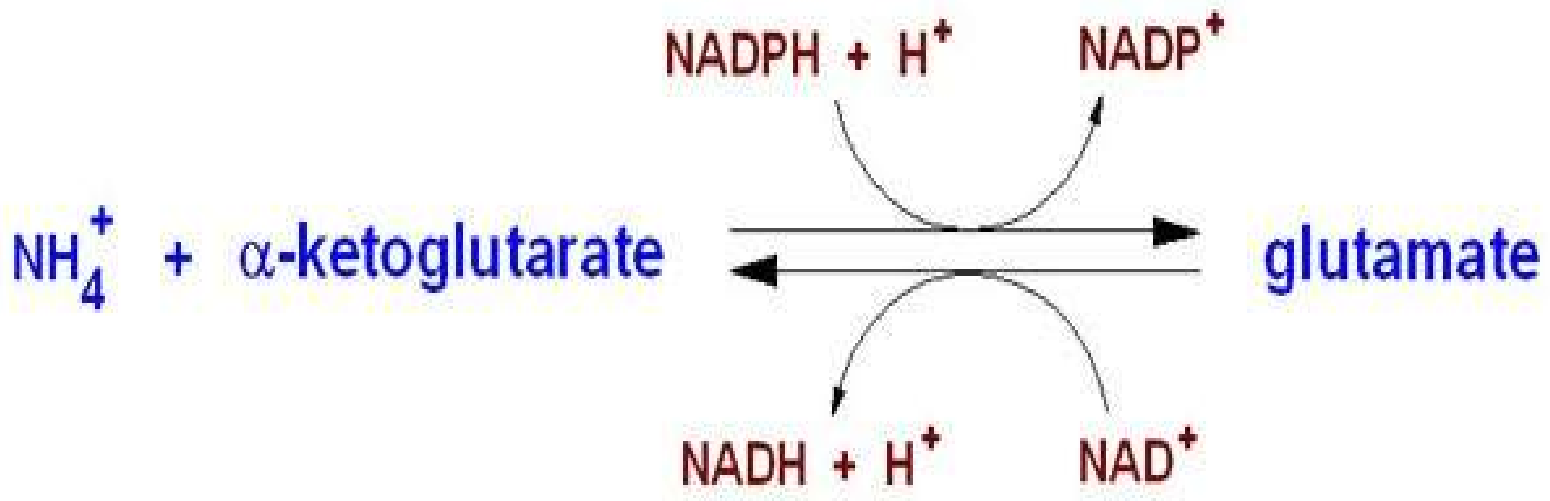



It is one of the few enzymes that can use **$NAD^+$ or $NADP^+$** as $e^-$ acceptor.

Oxidation at the $\alpha$-carbon is followed by hydrolysis, releasing $NH_4^+$.

# Glutamate Dehydrogenase

- Also glutamate dehydrogenase is important in converting free ammonia and α-KG to glutamate, forming one of the 20 amino acids required for protein synthesis



# Glutamate Dehydrogenase

The multiple roles of glutamate in nitrogen balance make it a gateway between free ammonia and the amino groups of most amino acids.

# GDH (Key Points)

GDH is a mitochondrial enzyme located in matrix, present in liver cells and most tissues.

Important for the following reasons:

- Is the principal source of $NH_4^+$ in the body.
- Link between TCA cycle & metabolism of AA ($\alpha$ keto acids are TCA cycle intermediates).
- In mammals, only reaction in which an inorganic molecule ($NH_4^+$) can be fixed to a C skeleton.
- GDH is the major AA oxidative pathway

# Glutamine synthetase

- Also important in several respects.
- Produces glutamine, one of the 20 major amino acids, the major amino acid found in the circulatory system.
- Role there is to carry ammonia to and from various tissues principally from peripheral tissues to the kidney, where the amide nitrogen of glutamine is hydrolyzed by the enzyme *glutaminase*
- The process regenerates **glutamate** and free ammonium ion, which is excreted in the urine

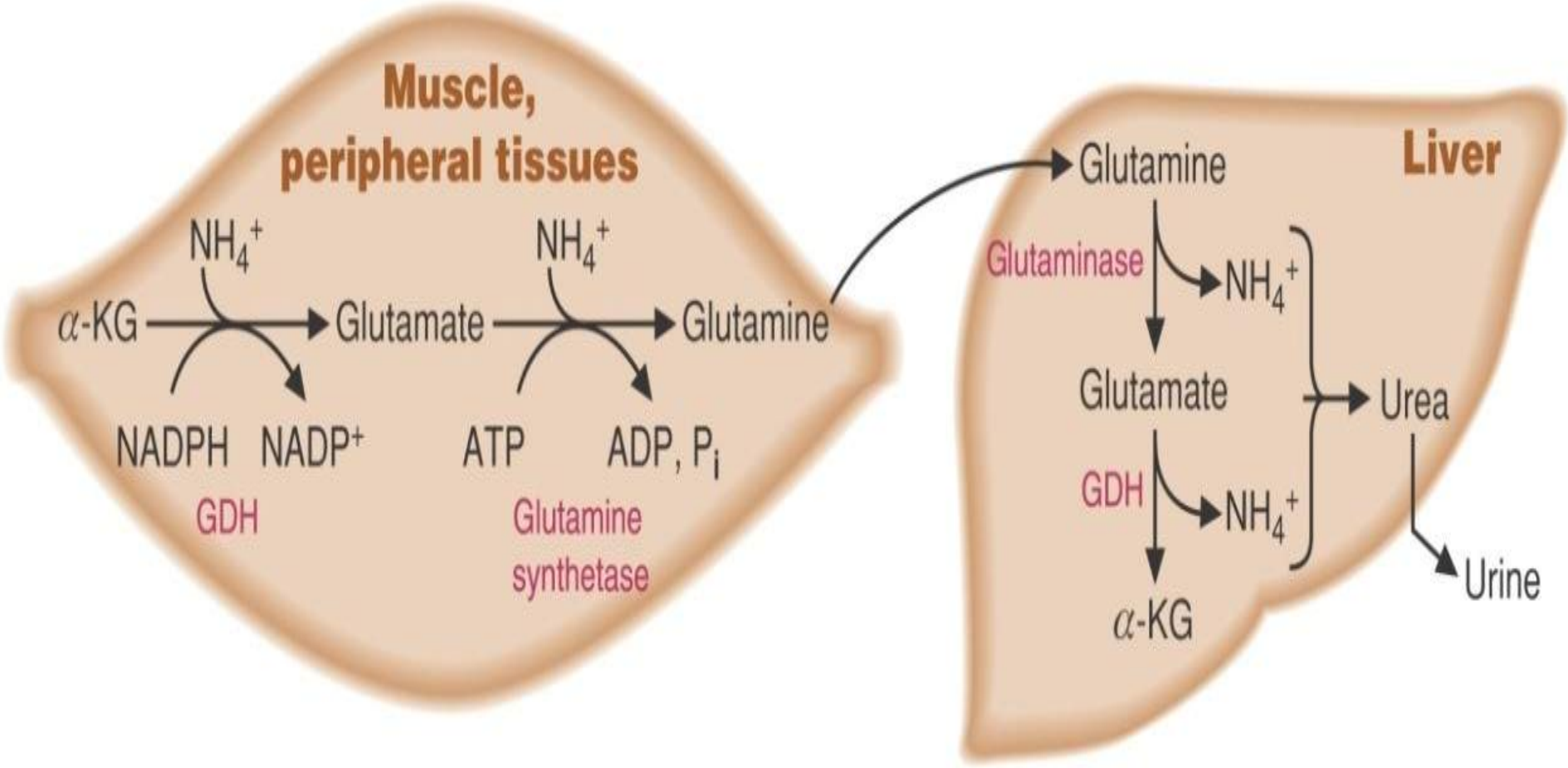



glutamine

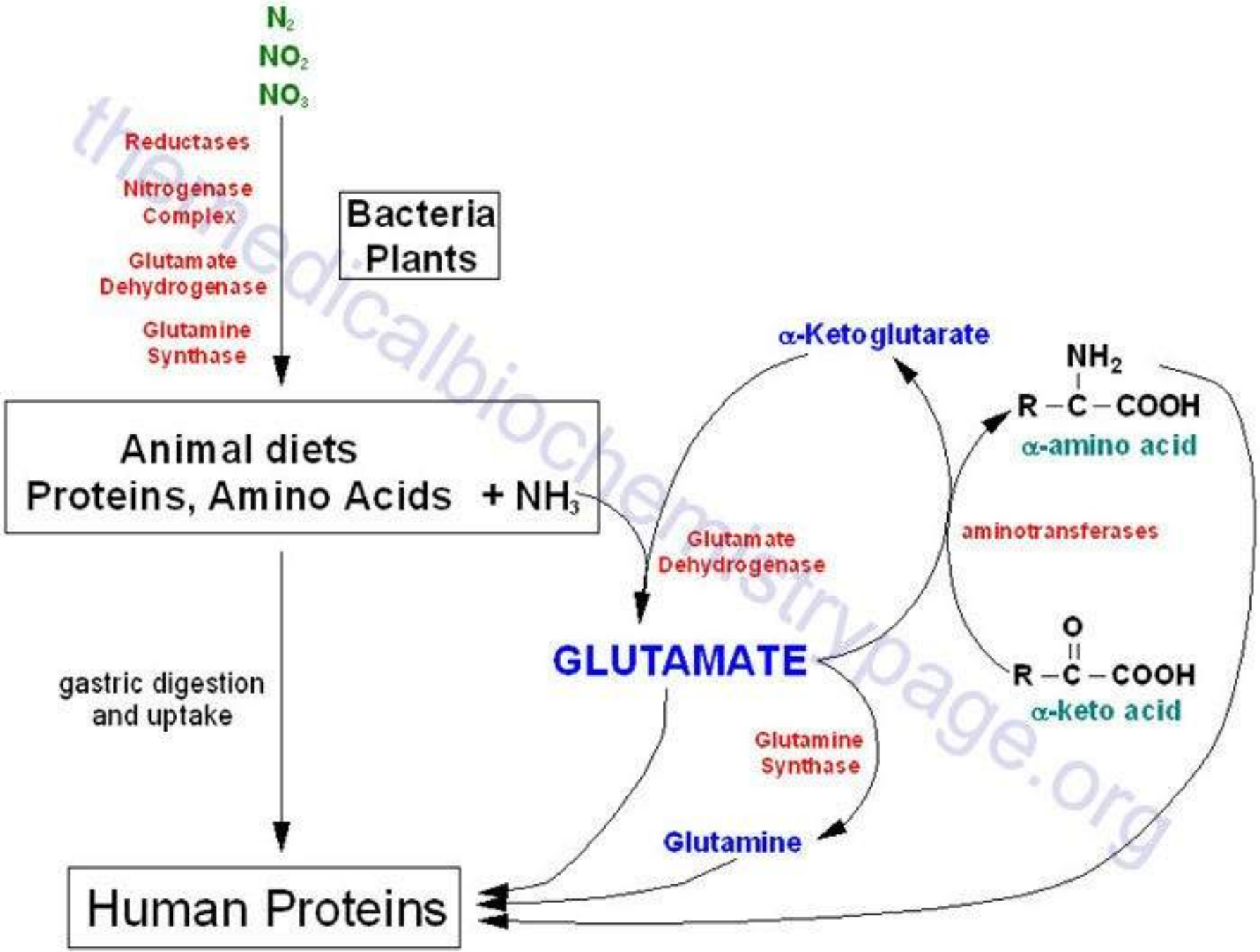
glutamate

## Glutamine takes 2 $NH_4$+ to liver

- Glutamine synthase forms glutamine
- Glutaminase removes $NH_4$+ → glutamate




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# Ammonia Disposal

- $NH_4^+$ must be disposed of due to toxicity

  symptoms = feeding intolerance, vomiting, lethargy. Irritability, respiratory distress, seizures and coma

- Most is disposed off in the form of **urea**

- Smaller amounts can be disposed of through the kidney via GLN

$GLN + H_2O \rightarrow GLU + NH_3 + H^+$ (glutaminase)

# Urea cycle in liver excretes toxic $NH_4^+$

# NITROGEN BALANCE

- Nitrogen balance occurs when the amount of nitrogen consumed equals that of the nitrogen excreted in the urine, sweat, and feces.
- Most healthy adults are normally in nitrogen balance.

# POSITIVE NITROGEN BALANCE

- This occurs when nitrogen intake exceeds nitrogen excretion.
- It is observed during situations in which tissue growth occurs, for example, in childhood, pregnancy, or during recovery from an emaciating illness.

# NEGATIVE NITROGEN BALANCE

- This occurs when nitrogen loss is greater than nitrogen intake.
- It is associated with inadequate dietary protein, lack of an essential amino acid, or during physiologic stresses, such as trauma, burns, illness, or surgery.

- The amount of dietary protein required in the diet varies with its biologic value.
- The greater the proportion of animal protein included in the diet, the less protein is required.
- The RDA for protein is computed for proteins of mixed biologic value at 0.8 g/kg of body weight for adults, or about 56 g of protein for a 70-kg individual.

- People who exercise strenuously on a regular basis may benefit from extra protein to maintain muscle mass a daily intake of about 1 g/kg has been recommended for athletes.
- Women who are pregnant or lactating require up to 30 g/day in addition to their basal requirements.
- To support growth, children should consume 2 g/kg/day.

# CONSUMPTION OF EXCESS PROTEINS

- There is no physiologic advantage to the consumption of more protein than the RDA.
- Protein consumed in excess of the body's needs is deaminated, and the resulting carbon skeletons are metabolized to provide energy or acetyl coenzyme A for fatty acid synthesis.
- When excess protein is eliminated from the body as urinary nitrogen, it is often accompanied by increased urinary calcium, increasing the risk of nephrolithiasis and osteoporosis.

# PROTEIN SPARING EFFECTS OF CARBOHYDRATES

- The dietary protein requirement is influenced by the carbohydrate content of the diet.
- When the intake of carbohydrates is low, amino acids are deaminated to provide carbon skeletons for the synthesis of glucose that is needed as a fuel by the central nervous system.

- If carbohydrate intake is less than 130 g/day, substantial amounts of protein are metabolized to provide precursors for gluconeogenesis.
- Therefore, carbohydrate is considered to be “protein-sparing,” because it allows amino acids to be used for repair and maintenance of tissue protein rather than for gluconeogenesis.

# Nitrogen balance

-equilibrium

N in = N out

healthy adult

-negative nitrogen balance

N in < N out

illness, starvation

-positive nitrogen balance

N in > N out

-person recovering from illness

-growing child

# Regulation of amino acid and protein metabolism

-demand for amino acid and protein anabolism -eg if need positive N balance or to keep in N equilibrium

-if lots of glucose available

-demand for amino acid and protein catabolism-eg if in negative N balance or to keep in N equilibrium

-if short of glucose


Thank You